

Sulfuric Acid 96% — Technical Grade

PRODUCT DATA SHEET
FT-H2SO496-2024

1. PRODUCT IDENTIFICATION

CAS No.	7664-93-9	EC No.	231-639-5
Formula	H2SO4	Molar mass	98.08 g/mol
Chemical family	Strong mineral acid	Grade	Technical

Available pack sizes: 25 L jerrican (HDPE) | 200 L drum (HDPE) | 1000 L tank (HDPE IBC)

2. PHYSICO-CHEMICAL PROPERTIES

Property	Value
Physical state	Oily liquid, colourless to slightly yellowish
Odour	Slightly acrid at high concentration
Boiling point	330 °C (with decomposition)
Freezing point	3 °C (H2SO4 96%)
Density at 20°C	1.835 g/mL
Viscosity at 20°C	21.4 mPa·s
pH (1% dilute solution)	< 1
Miscibility with water	Miscible (strongly exothermic)
Vapour pressure at 20°C	< 0.1 hPa
Flash point	Not applicable

3. ANALYTICAL SPECIFICATIONS

Parameter	Specification
H2SO4 content	95.0–97.0%
Ignition residue	<= 0.005%
Iron (Fe)	<= 5 ppm
Chlorides (Cl-)	<= 2 ppm
Arsenic (As)	<= 0.05 ppm
Lead (Pb)	<= 0.1 ppm
Nitrates (NO3-)	<= 5 ppm
Reducing agents (SO2)	<= 5 ppm

4. APPLICATIONS AND USES

- Pickling and surface treatment of metals (steel, copper, aluminium)
- Electrolyte in lead-acid batteries
- Synthesis of metal sulfates and alums
- Water treatment (pH correction)
- Textile industry (acid bath for dyeing)
- Regeneration of ion-exchange resins

5. HAZARD INFORMATION (CLP/GHS)

Code	Hazard statement
H290	May be corrosive to metals
H314	Causes severe skin burns and eye damage
H335	May cause respiratory irritation

6. PRECAUTIONARY STATEMENTS

Code	Precautionary statement
P234	Keep only in the original HDPE or glass packaging
P260	Do not breathe vapours or aerosols
P280	Wear full personal protective equipment
P301+P330+P331	IF SWALLOWED: rinse mouth. Do NOT induce vomiting
P303+P361+P353	IF ON SKIN: rinse immediately with plenty of water
P405+P501	Store locked up. Dispose of in accordance with local regulations

7. STORAGE AND SHELF LIFE

Store in a ventilated, acid-resistant area away from organic matter, combustibles, bases and oxidisers. Never add water to sulfuric acid (risk of violent splashing). Storage temperature: 5–35°C. Unlimited shelf life in closed packaging.

8. TRANSPORT AND REGULATIONS

UN 1830 | ADR class 8 | Packing group II | Tunnel code E

9. DETAILED APPLICATION NOTES

Material compatibility

Before use at scale, check compatibility with seals, pipework and containers. Give preference to materials tested and validated by ChemCorp Industries. Refer to the chemical compatibility tables available on our technical portal (portail.chemcorp.fr/compat). If in doubt, contact our Technical Application service on +33 4 72 XX XX XX.

Quality control on receipt

On receipt, check the integrity of the packaging and that the batch numbers match the accompanying certificate of analysis. Carry out a density and pH check where available. In the event of an anomaly, keep the original packaging and contact our quality department within 48 hours. Procedure ref.: QC-RECEP-003.

Waste management

Waste from this product must be managed in accordance with local hazardous waste regulations (Directive 2008/98/EC). Contact an approved contractor for disposal. European Waste Code (EWC): 06 01 04* for acid solutions, 07 01 01* for halogenated solvents, 07 01 04* for other organic solvents. Never discharge into drains or watercourses.

First aid (operational summary)

INHALATION: Move the person to fresh air, in a semi-seated position. If breathing is difficult, administer O₂ and call the emergency services (15 in France). SKIN CONTACT: Remove contaminated clothing. Rinse with plenty of water for at least 15 minutes. EYE CONTACT: Rinse with water for at least 15 minutes, holding the eyelids open. Consult an ophthalmologist. INGESTION: Do not induce vomiting. Rinse the mouth. Call the Poison Control Centre (European number: 0800 59 59 59).

Environmental data (in-house results)

BOD₅ (5 days): tested per ISO 5815. COD: tested per ISO 15705. Log K_{ow} (n-octanol/water partition coefficient): measured per OECD 117. Aerobic biodegradability: tested per OECD 301B. Aquatic toxicity (Daphnia magna): EC₅₀ per OECD 202. For the complete numerical values, refer to section 12 of the Safety Data Sheet.

Additional regulatory information

This product is registered under REACH: registration number available on request. It is / is not on the SVHC candidate list (last ECHA update: Jan 2025). IED Directive: applicable to installations using this product in quantities above 200 t/year. French ICPE: check the classification under the ICPE nomenclature (heading 1175, 4110 or 4XXX depending on the product). Certifications: ISO 9001:2015 | Compliant with Directive 2008/98/EC.